Transmission line fault detection: ML results

1500 simulated runs | 220 kV, 60 Hz, 40 km line | 37 features from one cycle of V and I at the sending end

Fault type (same-distribution test)

100.0%

0 errors in 60 per class: error likely < 5%

Location MAE (same-distribution test)

1.10 km

2.7% of line length, 85% within 2 km

Fault type, extrapolating to Rf > 10 ohm

95.5%

trained on Rf <= 10 ohm only

Location MAE, extrapolating to Rf > 10 ohm

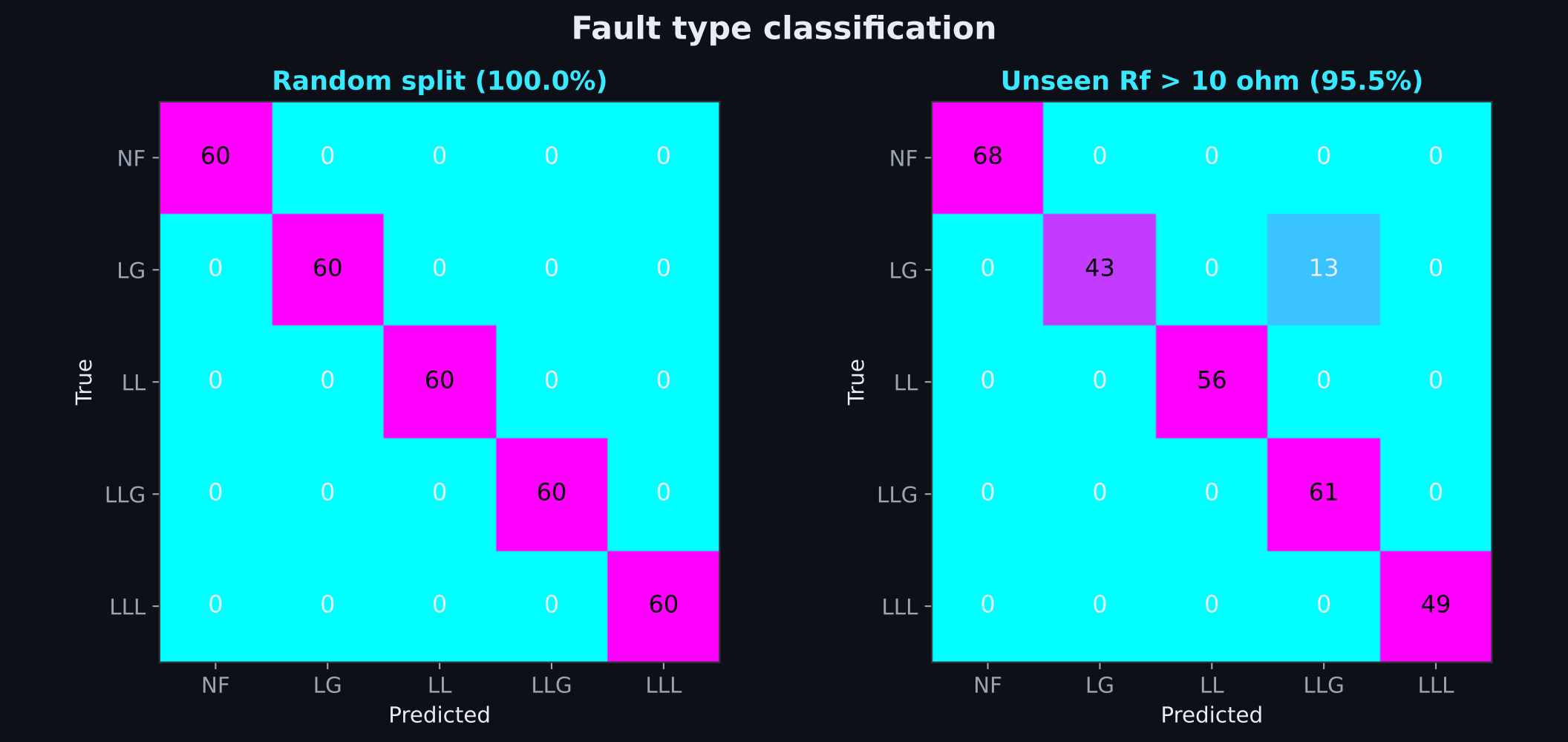
11.4 km

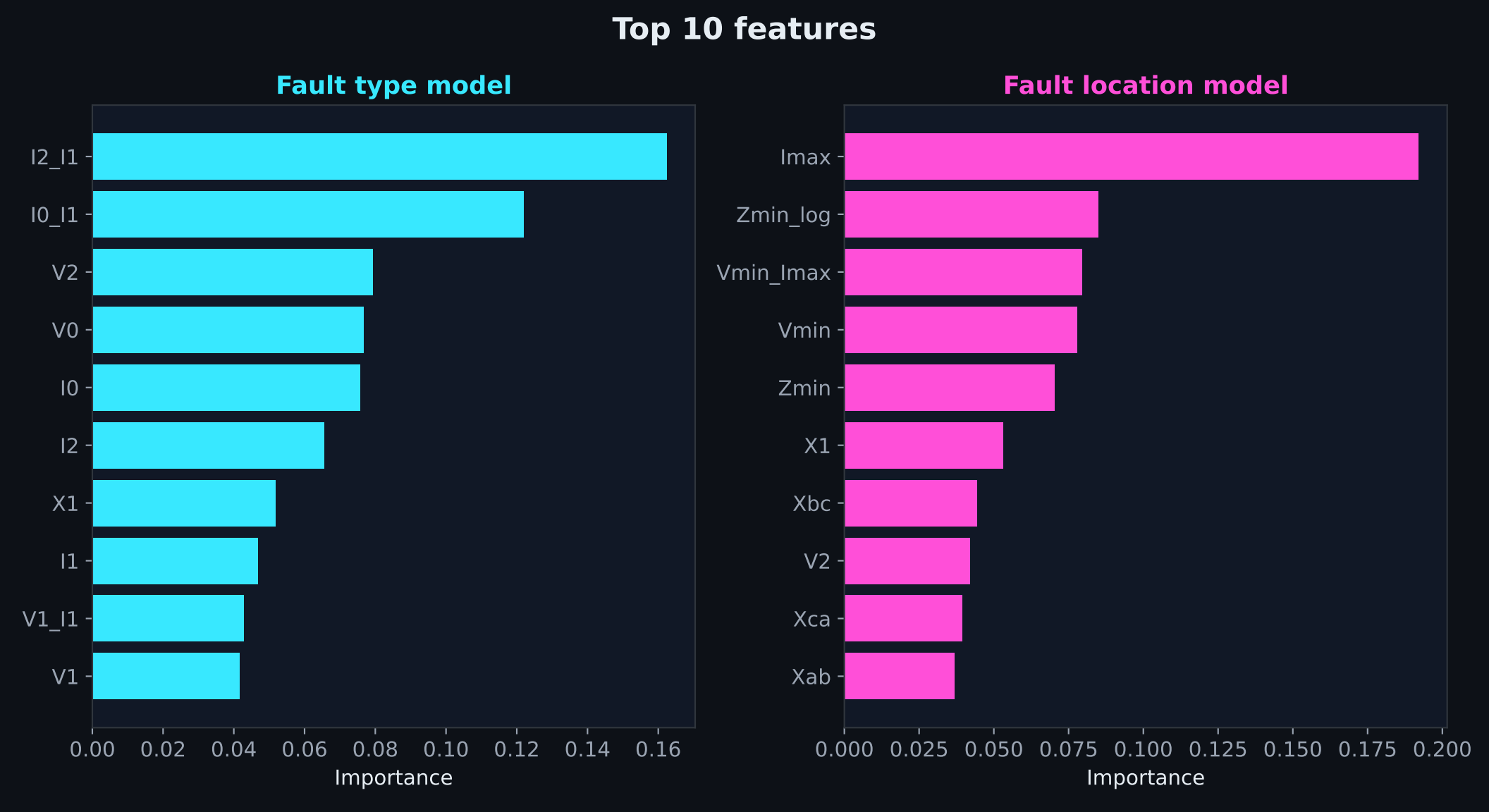
trained on Rf <= 10 ohm only

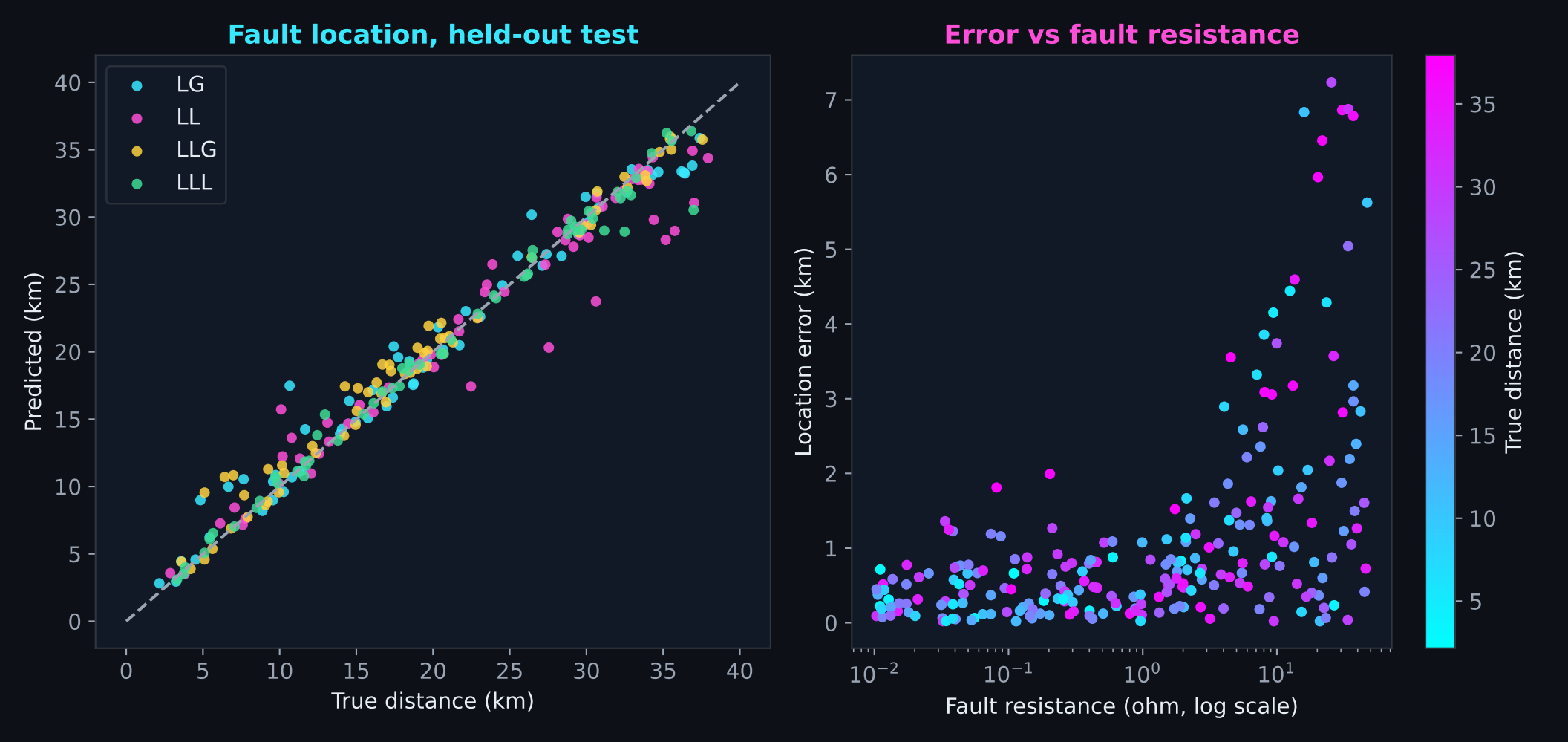
5-fold cross-validation accuracy: 100.0% +/- 0.0% | Decision Tree baseline: 99.7%

Unseen 3-10 ohm band: type accuracy 100.0%, location MAE 2.43 km | Extrapolation (Rf > 10): location MAE 11.4 km

Trigger: 0/1200 faults missed, 0/300 false alarms, median detection delay 1.4 ms


Fault type classification
Random split (100.0%)
Unseen Rf > 10 ohm (95.5%)
True
Predicted
NF
LG
LL
LLG
LLL
60 0 0 0 0
0 60 0 0 0
0 0 60 0 0
0 0 0 60 0
0 0 0 0 60
68 0 0 0 0
0 43 0 13 0
0 0 56 0 0
0 0 0 61 0
0 0 0 0 49


Top 10 features
Fault type model
I2_I1
I0_I1
V2
V0
I0
I2
X1
I1
V1_I1
V1
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
Importance
Fault location model
Imax
Zmin_log
Vmin_Imax
Vmin
Zmin
X1
Xbc
V2
Xca
Xab
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Importance


Fault location, held-out test
LG
LL
LLG
LLL
Predicted (km)
True distance (km)
0
5
10
15
20
25
30
35
40
Error vs fault resistance
Location error (km)
Fault resistance (ohm, log scale)
10^{-2}
10^{-1}
10^{0}
10^{1}
0
1
2
3
4
5
6
7
True distance (km)
5
10
15
20
25
30
35

Why the classes are separable

